

Bridge Across Three Structures: Connes-Moscovici, Yang-Mills/Sevostyanov, and the Riemann Hypothesis

Overview: What Is Being Connected

Three bodies of mathematics are in play. The first is the Connes-Moscovici 1982 L^2 -index theorem: for a G -invariant elliptic operator P on a homogeneous space $M = G/H$, the analytical index $\text{ind}_G P$, defined via a canonical von Neumann trace on the commutant of the left regular representation, equals a topological index computed from characteristic classes in relative Lie algebra cohomology. The second is Sevostyanov's non-perturbative quantization program: the Yang-Mills Hamiltonian is identified as a gradient-squared potential (with the Chern-Simons functional as the potential), and this structure allows a canonical family of quantizations on weighted L^2 spaces, producing, in the abelian case, a rigorous mass gap via the Bochner-Minlos theorem and Fock space spectral theory. The third is the narrative in file:18: the zeta function, after a cumulative-phase time change ψ , becomes a band-limited stationary process in the Paley-Wiener space $\text{PW}_{[-2,0]}$, from which the Riemann hypothesis and simplicity of zeros are argued via Hardy space theory, Schwarz reflection, and Bulinskaya's theorem.^{[1][2][3][4]}

The connections are not metaphorical. They run along eight identifiable mathematical axes, each of which is a structural correspondence — same theorem, same construction, same obstruction — realized in different guises across the three texts. These are documented in full below.

Connection I: The Von Neumann Trace as Quantization Measure

In Connes-Moscovici, the entire index theory rests on the G -trace tr_G , the canonical semifinite faithful normal trace on the von Neumann algebra $\mathcal{M}_G$ — the commutant of the left regular representation L of G on $L^2(G)$. Its defining property is^[^4]

$$\text{tr}_G(R(f)^* R(f)) = \int_G |f(g)|^2 dg.$$

This trace replaces ordinary dimension: instead of $\dim \ker P$, one computes $\dim_G \ker P = \text{tr}_G K_{\ker P}$, and this real number is finite even when the kernel is infinite-dimensional in the ordinary sense. The key computational identity (Corollary 2.3 in CM) is $\text{tr}_G(P) = k_P(1)$ — the G -trace of a G -invariant operator equals the diagonal value of its convolution kernel at the identity.^[^4]

In Sevostyanov, the Hilbert space on which the quantized Yang-Mills Hamiltonian acts is $L^2(\mathcal{D}/\mathcal{K}, d\mu)$, where μ is a probability measure with "density" $\psi e^{-2\phi}$ (with $\phi = CS$ the Chern-Simons functional). This measure is not Lebesgue — it is a weighted measure determined by the potential, exactly as in a ground-state representation. The choice of ψ is the *quantization ambiguity*: different choices produce unitarily inequivalent quantum theories. After the unitary intertwining trick $U : L^2(M, d\mu) \rightarrow L^2(M, d\text{vol})$, $f \mapsto f \cdot \sqrt{\psi e^{-2\phi}}$, the Hamiltonian becomes^{[2][3]}

$$\hat{H} = \frac{1}{2} \sum_a \xi_a^* \xi_a : L^2(M, d\mu) \rightarrow L^2(M, d\mu),$$

where ξ_a is directional differentiation and ξ_a^* its adjoint in the *weighted* L^2 . The measure $d\mu$ plays the role of the ground state wave function: the potential information is encoded in the measure rather than in the operator, exactly as in the von Neumann trace picture in CM where the G-trace encodes the Haar measure.[^3]

In the zeta narrative, the *covariance kernel* $K(t, s) = \sqrt{\psi'(t)} \cdot \text{sinc}(\psi(t) - \psi(s)) \cdot \sqrt{\psi'(s)}$ defines a positive semidefinite operator C on $L^2[0, \infty)$ [^1]. The diagonal $K(t, t) = 2\psi'(t) = 2\theta'(t)|\zeta(\frac{1}{2} + it)|^2$ is the reproducing kernel evaluation at the diagonal — the analogue of $k_P(1)$ in CM. All three constructions share the same structural role for their respective "measures" or "traces": they encode the geometry of the action (Haar measure, ground-state weight, or spectral density) and turn a potentially infinite-dimensional kernel into a finite (or controlled) quantity.

Connection II: Orbit Spaces, Averaging, and Symplectic Reduction

The configuration space in all three theories is a quotient of a function space by a group.

In CM, $M = G/H$ is the space of left cosets, and every construction passes through the *averaging operation* $\text{Av}(P) = \int_G L(g)PL(g)^{-1} dg$, which converts an ordinary compactly supported Ψ DO into a G -invariant one. This averaging is the fundamental reduction step: it projects from the "over-parameterized" space of all operators down to the G -equivariant sector. Lemma 1.2 shows that every G -invariant Ψ DO is an average: $P = \text{Av}(fP)$ for any cut-off function f with $\int f dg = 1$. [^4]

In Sevostyanov, the Marsden-Weinstein symplectic reduction theorem (Proposition 3) is applied to the gauge group $\mathcal{K}$ acting on the phase space $T\mathcal{D}$. The moment map is $\mu(A, E) = \text{div}_A E$ (Gauss's law), the constraint surface is $\{E : \text{div}_A E = 0\}$, and the reduced phase space is $\mu^{-1}(0)/\mathcal{K} \cong T^*(\mathcal{D}/\mathcal{K})$. The horizontal subspaces $\ker \text{div}_A$ (Coulomb gauge condition) are the analogues of CM's homogeneous tangent directions. Lemma 1 in Sevostyanov proves that the annihilator of the gauge orbit inside $T\mathcal{D}$ equals the kernel of the moment map's differential — this is the exact analogue of CM's identification of the symbol as the element of $K_H(V)$, the equivariant K-theory group of the co-isotropy representation. [2][3]

In the zeta narrative, the *time change* $\psi(t) = \int_0^t \theta'(u)|\zeta(\frac{1}{2} + iu)|^2 du$ is the reduction operation [^1]: it absorbs the non-stationarity of the zeta process (the irregular spacing of zeros, the growth of $\theta'(t)$) into a single reparametrization, yielding a stationary band-limited process $X(T)$ on the quotient timeline. This is a Priestley oscillatory-process reduction: the evolutionary spectrum is collapsed into a stationary spectrum by the time change, just as CM's averaging collapses a general Ψ DO into a G -invariant one.

The Gribov obstruction in Sevostyanov — the fact that $\mathcal{D}/\mathcal{K}$ cannot be globally section-ized — is the infinite-dimensional analogue of CM's non-existence of a global right-inverse for the projection $G \rightarrow G/H$ when the isotropy representation is non-trivial. Both are topological obstructions to globally choosing representatives; both are handled by working in the quotient category throughout, rather than fixing a gauge. [^2]

Connection III: The Index as a Topological Invariant of the Spectrum

The deepest structural connection runs through the passage from *analytical* to *topological* index.

In CM, the main theorem $\text{ind}_G P = \text{ind}_t[\sigma(P)]$ says: the real number $\dim_G \ker P - \dim_G \ker P^*$, which is an analytic datum (count L^2 -solutions), equals a characteristic class integral:[^4]

$$\text{ind}_G D_\varepsilon^+ = (-1)^n (\text{ch}(\varepsilon) \cdot \hat{A}(\mathfrak{g}, H)) [V],$$

where $\text{ch}(\varepsilon)$ is the Chern character of the representation ε of H , and $\hat{A}(\mathfrak{g}, H)$ is the relative $\hat{A}$ -class computed from the curvature of the canonical connection on $G \rightarrow G/H$. The topological index depends only on the principal symbol $\sigma(P)$ — not on the specific operator — and is invariant under homotopy of the symbol.

In Sevostyanov, the *key structural observation* (Proposition 5, the entire pivot of the paper) is that the potential vector field $G = *F$ is the gradient of the Chern-Simons functional $CS: G = \text{grad } CS$ [2]. Therefore the Yang-Mills potential $\frac{1}{2} \langle G, G \rangle = \frac{1}{2} \|\text{grad } CS\|^2$. The Hamiltonian is of the form

$$H = \frac{1}{2} (\langle p, p \rangle + \|\text{grad } CS\|^2).$$

This is a gradient-squared Hamiltonian: the same structure as a Witten-type Morse theory, where the "potential function" is a topological invariant (the Chern-Simons functional) and the Hamiltonian measures proximity to the critical locus (flat connections). The quantization measure

$d\mu \propto e^{-2CS(A)} dA$ is the analogue of the Chern character: it transforms a topological datum (the winding number and homotopy type of connections) into an analytic datum (the spectrum of $\hat{H}$). The mass gap $\frac{1}{2}m$ is the "topological index" of the quantization: it is parametric, but it arises from the topological structure of the Chern-Simons functional, not from a specific potential well imposed by hand.[^3]

In the zeta narrative, the Paley-Wiener space $PW_{[-2,0]}$ is itself a topological invariant: it is the reproducing kernel Hilbert space associated to the spectral support $[-2, 0]$, which is determined entirely by the structure of the Riemann-Siegel expansion (the ratio $\log n / \theta'(t)$ after the time change). The band-limitation $\hat{X}(\omega) = 0$ for $\omega > 0$ is an index condition: it says the Hardy-space index of X in the upper half-plane is $-\infty$ (no poles above the real axis), which is the spectral manifestation of zero-free half-planes. The Fourier support $[-2, 0]$ plays the role of the topological index: it encodes the analytic information about the zeros globally, just as the Chern character integral encodes the index.[^1]

Connection IV: The Chern-Simons Functional as Linking Thread

The Chern-Simons functional appears in two of the three papers explicitly and underlies the third implicitly.

In CM, the topological index formula involves the Chern character $\text{ch}(\varepsilon) = [\text{tr exp}(\Omega_\varepsilon)]$, computed from the curvature $\Omega_\varepsilon(X, Y) = i\varepsilon(\Omega(X, Y))$ of the canonical connection on the homogeneous bundle. The Chern-Weil representatives are evaluated in the relative Lie algebra cohomology $H^*(\mathfrak{g}, H; \mathbb{R})$, which is computed from the *same* data as the Chern-Simons form: the projection $\theta: \mathfrak{g} \rightarrow \mathfrak{h}$ and the curvature $\Omega = -\frac{1}{2}[-, -]_{\mathfrak{h}}$. [^4]

In Sevostyanov, the Chern-Simons functional $CS(A) = \int_{\mathbb{R}^3} (A \wedge dA + \frac{1}{3} [A \wedge A] \wedge A)$ is the *potential* whose gradient generates the Yang-Mills magnetic field. In the abelian case, $CS(A) = \frac{1}{2} \int A \wedge dA$ (magnetic helicity), and the exponent of the Gaussian measure is

$$-\frac{1}{2c^2} \langle G, G \rangle - 2 CS(A) - \frac{1}{2} (c^2 + m) \langle A, A \rangle,$$

which combines the topological invariant CS with a regularization quadratic form. The measure is entirely determined by the Chern-Simons structure.[^3]

In the zeta narrative, the "topological" content is carried by the Riemann-Siegel phase $\theta(t) = \text{Im} \log \Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2} \log \pi$ [^1]. This phase is the arithmetic analogue of the Chern-Simons phase: it is a winding number of the Γ -factor of ζ , and the time change $\psi(t) = \int_0^t \theta'(u) |\zeta(\frac{1}{2} + iu)|^2 du$ is built from it. The band-limitation argument shows that the instantaneous frequency $\omega_n(T) = \log n / \psi'(\psi^{-1}(T))$ of each Riemann-Siegel term is ≤ 2 after the time change — a bound that comes from the same growth rate of $\theta'(t) \sim \frac{1}{2} \log(t/2\pi)$ that controls the Chern-Simons phase winding. The Chern-Simons functional, in gauge theory, measures the "topological charge" of a field configuration; the Riemann-Siegel phase, in number theory, measures the topological charge of the Γ -factor — both are the "winding" that the respective principal objects carry.

Connection V: Self-Adjointness via Three Different Mechanisms

All three theories arrive at self-adjointness as the organizing structural requirement, but via distinct mechanisms that are mathematically equivalent in spirit.

In CM, self-adjointness of the relevant operators is automatic: G -invariant Ψ DOs of order zero are in the commutant $\mathcal{M}_G(\xi)$ of the left regular representation, which is a von Neumann algebra and hence consists of bounded operators. The trace tr_G is faithful and normal on this algebra, so the spectral theory is entirely internal to the algebra. The key consequence is the Proposition 1.7 exact sequence $0 \rightarrow C_G^*(\xi) \rightarrow \Psi_G^*(\xi) \rightarrow C_H(S, E, E) \rightarrow 0$, which ensures the symbol map is surjective and the quotient is Abelian — a C^* -algebraic self-adjointness statement.[^4]

In Sevostyanov, self-adjointness is not automatic: the quantized Hamiltonian

$$\hat{H} = \frac{1}{2} \sum_a \xi_a^*(x) \xi_a(x)$$

is initially defined only as a non-negative symmetric operator on smooth compactly supported functions in $L^2(M, d\mu)$. The Friedrichs extension theorem then guarantees the existence of a canonical self-adjoint extension: since the bilinear form $Q(f, f) = \frac{1}{2} \sum_a \|\xi_a f\|^2 \geq 0$ is non-negative and densely defined, it is closable, and its closure defines the maximal self-adjoint extension [^2]. The condition $1 \in L^2(M, d\mu)$ (the measure is finite = probability measure) ensures $\hat{H}(1) = 0$, confirming the vacuum as the ground state with eigenvalue 0.

In the zeta narrative, self-adjointness of the Hilbert-Pólya operator $D = -i\psi'(t)^{-1} \cdot d/dt$ on H_K is established by Stone's theorem: the translation group $V_\epsilon f(t) = f(\psi^{-1}(\psi(t) + \epsilon))$ is a strongly continuous unitary group on H_K (it is translation on $\text{PW}_{[-2,0]}$ conjugated by the unitary time-change isomorphism), so its generator is self-adjoint. The spectrum of D is $[-2, 0]$, absolutely continuous — the direct operator-theory translation of band-limitation. The Riemann zeros appear

not as eigenvalues but as singularities of the coefficient $1/\psi'(t_n)$: points where the domain of D is constrained, forcing every function in $\text{Dom}(D)$ to vanish at t_n .^[1]

The three mechanisms — von Neumann algebra structure, Friedrichs extension of a non-negative form, Stone's theorem applied to a unitary group — are the three fundamental ways that self-adjointness arises in functional analysis. Each is the appropriate mechanism for its context (bounded operator in a C^* -algebra; semibounded differential operator; translation group on a function space), and all three express the same underlying principle: the operator's spectrum is real because the underlying geometry is Hermitian.

Connection VI: The Kernel of the Operator = The Discrete Geometry

In all three theories, the kernel of the central operator encodes the discrete arithmetic or geometric data of the problem.

In CM, Theorem 6.1 states that $\ker P$ is a finite direct sum of irreducible discrete series representations of G . The discrete series are the irreducible unitary representations with L^2 -matrix coefficients — they are the representations that "fit" in $L^2(G)$ without continuous spectral contribution. The existence of discrete series is tied to the rank condition: G has discrete series if and only if it has a compact Cartan subgroup ($\text{rank}(K) = \text{rank}(G)$, Harish-Chandra's theorem). The corollary is a vanishing theorem: if $\text{rank}(K_S) < \text{rank}(S)$, then $\ker P = \{0\}$ for all elliptic P — no nontrivial L^2 solutions exist.^[4]

In Sevostyanov, the kernel of $\hat{H}$ is one-dimensional, spanned by the vacuum state $\Omega = 1 \in L^2(\mathcal{D}/\mathcal{K}, d\mu)$. The eigenvalue 0 is simple (Theorem in §4.3). The mass gap $\frac{1}{2}m$ is the infimum of the spectrum above 0: no state between 0 and $\frac{1}{2}m$ exists. In the Fock space picture, every one-particle state has energy $\geq \frac{1}{2}m$ because the one-particle energies are $\frac{1}{2}(|\xi|^2 + m)^{1/2} \geq \frac{1}{2}m^{1/2}$ (in the abelian case). The vacuum is the unique element of $\ker \hat{H}$, and the mass gap is the gap between $\ker \hat{H}$ and the rest of the spectrum ^[2].

In the zeta narrative, the kernel of the covariance operator C is characterized by its *degeneracy locus*: $(Cf)(t_n) = 0$ for every Riemann zero t_n and every f in the domain. Every function in the range of C is forced to vanish at every nontrivial zero of ζ on the critical line. This is not an eigenvalue condition ($Cf \neq 0$ generally), but a *nodal* condition: the geometric locus of zeros of ζ is the locus where the covariance degenerates. The Riemann zeros are to C what the discrete series are to the CM kernel: the discrete points encoding the arithmetic geometry.^[1]

The parallel is sharpened by Montgomery's pair correlation conjecture and Rudnick-Sarnak's work: the Riemann zeros (in the ψ -time) follow GUE statistics, the same statistics as eigenvalues of random matrices from $U(n)$, the same group whose representation theory generates the discrete series of semisimple groups. The "discrete series" in CM and the "discrete spectrum" of Riemann zeros in the RH narrative are dual perspectives on the same phenomenon.

Connection VII: Heat Kernel, Asymptotic Expansion, and the Mass Gap

The heat kernel method is the technical bridge between the analytical and topological data in CM, and its structure recurs in both Sevostyanov and the zeta story.

In CM, the proof of $\text{ind}_G P = \text{ind}_t[\sigma(P)]$ proceeds via the McKean-Singer identity with the G -trace: [^4]

$$\text{tr}_G e^{-t\Delta^+} - \text{tr}_G e^{-t\Delta^-} = \dim_G \ker D^+ - \dim_G \ker D^- = \text{ind}_G D^+$$

for all $t > 0$. The left side has an asymptotic expansion as $t \rightarrow 0^+$:

$$\text{tr}_G e^{-t\Delta^\pm} \sim \sum_{k \geq -m} c_k(\Delta^\pm) t^{k/2},$$

whose constant term $c_0(\Delta^+) - c_0(\Delta^-)$ gives the topological index via the Chern-Weil integral. The index is the t -independent part of the heat trace difference.[^4]

In Sevostyanov, the harmonic oscillator example makes the heat-kernel connection explicit. The quantized Hamiltonian with $\psi = e^{-x^2/2}$ is the Hermite operator

$$\hat{H} = -\frac{1}{2} e^{x^2/2} \frac{d}{dx} e^{-x^2/2} \frac{d}{dx} : L^2(\mathbb{R}, e^{-x^2/2} dx) \rightarrow L^2(\mathbb{R}, e^{-x^2/2} dx),$$

with spectrum $\{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$. The heat semigroup $e^{-t\hat{H}}$ is the Mehler kernel (Ornstein-Uhlenbeck heat kernel), and its trace $\text{tr}(e^{-t\hat{H}}) = \sum_{n \geq 0} e^{-nt/2} = (1 - e^{-t/2})^{-1}$ generates the Bernoulli numbers via Taylor expansion — the same Bernoulli numbers that appear in the $\hat{A}$ -genus formula and in the Chern character. This is not a coincidence: both the harmonic oscillator heat trace and the CM index formula are instances of the same regularized determinant $\det(1 - e^{-t\hat{H}})$ appearing in two different avatars.[^2][3]

In the zeta narrative, the prolate spheroidal wave functions (PSWFs) are the eigenfunctions of the restricted covariance operator C_L on $[0, L]$. The PSWF eigenvalues $\lambda_n(L)$ are the heat-kernel eigenvalues of a truncated Paley-Wiener operator, and they exhibit a Landau-Widom phase transition near the Shannon number $2L/\pi$: eigenvalues near 1 for $n \ll 2L/\pi$ and near 0 for $n \gg 2L/\pi$. This is the zeta analogue of the heat-trace asymptotics in CM: the "index density" (transition width of the PSWF eigenvalue distribution) is the analogue of the $t \rightarrow 0$ constant term in the heat trace.[^1]

Connection VIII: The Killing Form, the Metric, and the Positivity Condition

The session narrative (file:18) opens with a problem in the differential geometry of Lie groups: for $G_{\mathbb{C}} = SL(n, \mathbb{C})$ with Cartan decomposition $G_{\mathbb{C}} = P \cdot G$, when does a smooth isometric section $f : (G, g) \rightarrow (G_{\mathbb{C}}, g_k)$ exist? The isometry condition reduces to:[^1]

$$B_k(v, w) + B_k(\varphi(v), \varphi(w)) = g_e(v, w), \quad v, w \in \mathfrak{su}(n),$$

which requires $g_e + B_k|_{\mathfrak{su}(n)}$ to be positive semidefinite. This positivity condition — the metric g must dominate the (negative of the) Killing form — is identified as structurally identical to Weil's positivity criterion for the Riemann hypothesis.

This Killing-form positivity is precisely the condition that enters CM's construction [^4]. The co-isotropy representation $V = \{\xi \in \mathfrak{g}^* : \xi|_{\mathfrak{h}} = 0\}$ carries an $\text{Ad}^*(H)$ -invariant metric (used to define the sphere $S = S(V)$ on which symbols are valued). The condition that the Bott class $\beta \in K_H(V)$ exists and generates $K_H(V)$ as an $R(H)$ -module requires precisely that the representation $\text{Ad}^* : H \rightarrow SO(V)$ be orientable — a positivity condition on the isotropy representation analogous to the positive semidefiniteness of $g_e + B_k$. When the Killing form on V is non-degenerate and the orientation is preserved, the Thom isomorphism $\tau : R(H) \rightarrow K_H(V)$ exists and the index formula applies.

In Sevostyanov, the positivity condition is the requirement that the exponent of the Gaussian measure be negative definite. The naive choice $\psi = \exp(-\frac{1}{2}\langle G, G \rangle)$ fails: the quadratic form in the exponent is indefinite (the cross term $-\langle G, G \rangle - 2CS(A)$ is not sign-definite). The correct ψ adds the extra term $-\frac{1}{2}(c^2 + m)\langle A, A \rangle$, and the resulting exponent[^3]

$$-\frac{1}{2c^2}\langle G, G \rangle - \frac{1}{2}(c^2 + m)\|A\|_{L^2}^2$$

is negative definite by a completing-the-square computation. This is the Sevostyanov analogue of the "metric must dominate the Killing form" condition: the quantization measure exists if and only if the quadratic form defining it is negative definite, i.e., the "metric" $\frac{1}{2}(c^2 + m)\|\cdot\|^2$ dominates the negative-definite-but-coupled term from the Chern-Simons cross term.

The RH narrative identifies this positivity condition with Weil's criterion: RH is equivalent to the Weil distribution $W(f * \bar{f}^\sharp) \leq 0$ for all admissible test functions f . The "metric" in this case is the L^2 norm of the test function, and the Weil quadratic form is the analogue of the Killing form — both are sesquilinear forms on a Lie-algebraic or function-analytic space, both require a global positive semidefiniteness to hold for the respective theorem to be true, and both have the same "infinitesimal-to-global" obstruction structure.[^1]

Connection IX: Non-Compactness as the Universal Obstruction, and How All Three Overcome It

The unifying technical obstruction in all three theories is *non-compactness*, and each theory has its own mechanism for taming it.

In CM, the non-compactness of $M = G/H$ means ordinary Fredholm theory fails: $\ker P$ can be infinite-dimensional in the naive sense. The resolution is the G-dimension: replace $\dim$ by $\dim_G = \text{tr}_G$, which is finite even for non-compact orbits. The cut-off function f (with $\int f dg = 1$) is the technical device: it localizes integrals over G without compact quotient, and the identity $\text{tr}_G(P) = k_P(1)$ (evaluation at the identity) is the "localization" of the infinite-volume trace.[^4]

In Sevostyanov, the non-compactness is twofold: the spatial domain is $\mathbb{R}^3$ (non-compact), and the configuration space $\mathcal{D}/\mathcal{K}$ is infinite-dimensional. The Bochner-Minlos theorem resolves both: it guarantees that a Gaussian measure on an infinite-dimensional nuclear space $\mathcal{S} \subset H \subset \mathcal{S}'$ exists (as a probability measure on the dual $\mathcal{S}'$) whenever the characteristic functional is positive definite, continuous, and normalized. The nuclear space $\mathcal{S}$ (Schwartz-class divergence-free vector fields) is the compact "core" that the measure lives on — the analogue of the compact quotient $\Gamma \backslash G$ in Atiyah's covering L^2 -index theorem. The spectral analysis of the covariance operator $\hat{C} = (-\Delta + m)^{-1}$ (spectrum contained in $(0, 1]$) confirms positivity and boundedness, validating the Bochner-Minlos construction.[^2][3]

In the zeta narrative, the non-compactness is the non-compact imaginary axis $\operatorname{Re}(s) = \frac{1}{2}$, $0 < \operatorname{Im}(s) < \infty$. The time change ψ converts the non-stationary process (growing height, irregular zero spacing) into a stationary one on all of $\mathbb{R}$. The Paley-Wiener theorem then controls the non-compactness: a function entire of exponential type with Fourier support in $[-2, 0]$ is polynomially bounded in every direction — the non-compactness of $\mathbb{R}$ is controlled by the exponential type, exactly as the G-trace controls the non-compactness of G in CM.[^1]

The Complete Map of Correspondences

Structure	CM (Connes-Moscovici 1982)	Sevostyanov (Yang-Mills)	RH Narrative (Zeta)	
Ambient space	$M = G/H$ homogeneous	$\mathcal{D}/\mathcal{K}$ gauge orbit	$\operatorname{Re}(s) = \frac{1}{2}$ critical line	
Group action	Left G on $L^2(M, \xi)$	Gauge group $\mathcal{K}$ on $\mathcal{D}$	$\mathbb{R}^\times$ reparametrization	
Reduction operation	Averaging $\operatorname{Av}(P)$	Marsden-Weinstein reduction	Time change $\psi(t)$	
Canonical measure/trace	tr_G on $\mathcal{M}_G$	$d\mu = \psi e^{-2\phi} d\operatorname{vol}$	Covariance kernel $K(t, s)$	
Trace localization	$\operatorname{tr}_G(P) = k_P(1)$	Weighted L^2 intertwining	Diagonal $K(t, t) = 2\psi'(t)$	
Topological invariant	Chern character $\operatorname{ch}(\varepsilon)$ in $H^*(\mathfrak{g}, H; \mathbb{R})$	Chern-Simons functional $CS(A)$	Riemann-Siegel phase $\theta(t)$	
Index / gap	$\operatorname{ind}_G P = (\operatorname{ch} \cdot \hat{A})[V]$	Mass gap $= \frac{1}{2} m$	Band support $[-2, 0]$	
Self-adjointness mechanism	Von Neumann algebra, C^* -extension	Friedrichs extension	Stone's theorem	
Kernel encodes	Discrete series representations	Vacuum state, $\ker \hat{H} = \mathbb{C} \cdot \Omega$	Zeros of ζ (nodal locus of C)	
Non-compactness tamed by	G-dimension $\dim_G = \operatorname{tr}_G K$	Bochner-Minlos on nuclear space	Paley-Wiener exponential type	
Positivity condition	$(g_e + B_k$	$\frac{\operatorname{su}(n)}{n} \geq 0$	Exponent $-\frac{1}{2c^2} \ G\ ^2 - \frac{1}{2}(c^2 + m) \ A\ ^2$ negative definite	Weil $W(f * \bar{f}^\sharp) \leq 0$
Vanishing criterion	$\operatorname{rank}(K) < \operatorname{rank}(G) \Rightarrow \ker P = 0$	Free theory ($m = 0$) has no gap	Off-critical-line zeros absent by Hardy space membership	
Heat / spectral structure	McKean-Singer: $\operatorname{tr}_G(e^{-t\Delta^+} - e^{-t\Delta^-}) = \operatorname{ind}_G$	Hermite operator heat trace $\sim$ Bernoulli numbers	PSWF eigenvalues, Landau-Widom transition	

The Synthesis: One Theorem in Three Languages

The governing principle, stated uniformly across all three, is:

A G -invariant elliptic operator on a non-compact space with group symmetry has an L^2 -index (or spectral gap, or band-limited zero structure) that is controlled by a topological invariant of the group action, constructed from the same Chern-Weil data that governs the representation theory. Self-adjointness forces the index/gap/zeros to be real; the von Neumann or Gaussian measure converts the non-compact geometry into a finite datum.

In CM, this theorem is the index formula. In Sevostyanov, it is the mass gap theorem for the abelian case, and the conjecture for the non-abelian case. In the RH narrative, it is the band-limitation claim and the Hardy-space argument.^{[3][2][1][4]}

The non-abelian Yang-Mills problem that Sevostyanov leaves open — constructing the non-Gaussian measure $d\mu \propto \psi e^{-2CS(A)} dA$ for a non-abelian gauge group — is exactly the problem that the CM theorem would address if it could be extended to infinite-dimensional G . In the CM framework, the formal degree $\deg(\pi)$ of a discrete series representation is the analogue of the mass m : it is a positive real number (non-integer in general) measuring the "density" of the representation in $L^2(G)$, just as m measures the spectral gap. The reciprocity theorem (CM Proposition 7.3A) — expressing $\text{ind}_G D$ as a sum $\sum_{\pi} (\dim \ker D_{\pi} - \dim \ker D_{\pi}^*) \deg(\pi)$ over discrete series — is the operator-algebraic analogue of the Wiener-Itô-Segal decomposition of the mass-gap Fock space into particle sectors.

The remaining open problem — whether the band-limitation of $X(T)$ is rigorously established under uniform control of the Riemann-Siegel remainder under the time change ψ — is the zeta analogue of the Sevostyanov open problem: both require constructing, rigorously, a specific "measure" or "spectral object" on an infinite-dimensional space (the space of connections in one case, the space of all Riemann-Siegel remainder terms in the other) and showing it has the required properties. Both have the same logical structure: abelian/local case is fully rigorous, non-abelian/global case requires controlling a non-Gaussian/non-uniform object that is the principal obstruction.

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4. [AC-biblio57.pdf](#) - Annals of Mathematics

The L2-Index Theorem for Homogeneous Spaces of Lie Groups

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